Lab 14: Microbiology — Known Strains, Plating, and Antibiotic Discs

BIOL-8

Name: ______________________ **Date:** ______________________

Overview

This lab uses **14 bacterial species** maintained on **tryptic soy agar (TSA)** and in **tryptic soy broth (TSB)** (including **agar slants**). You will **plate and spread** assigned strains, then place **antibiotic-impregnated discs** in labeled quadrants to observe **antibiotic susceptibility** after incubation. In **Part B**, each student uses **take-home (or classroom) air-exposure plates** to grow whatever microbes settle from the environment.

The skills you practice here — aseptic technique, forming and testing a hypothesis, and connecting molecular mechanisms to observable results — are foundational to clinical microbiology, food safety, and public-health laboratory work.

Interpretation after growth (zones, colony types, Gram logic vs. what you see) is completed in Lab 14 follow-up once plates have incubated.

Learning Objectives

- Work with **BSL-1** cultures using **aseptic technique** (sterile tools, minimal open time).
- **Prepare labeled Petri dishes** with **four quadrants** for multiple antibiotics on one lawn of a single strain.
- **Inoculate** from broth using a **sterile dropper** and **spread** for a uniform lawn before placing antibiotic discs.
- **Choose four antibiotics** from the reference list and use the **same four** across all three strain plates so comparisons are **controlled**.
- **Hypothesize** which antibiotics will be effective against each strain **before** plating, based on Gram status and drug targets.
- **Contrast Gram-positive and Gram-negative** cell envelopes (peptidoglycan thickness, outer membrane) and relate envelope structure to **why some antibiotic classes preferentially affect one category**.

- **Map** each disc to its **drug class** and **primary molecular target**.
- Complete the **air-exposure** plate protocol and **hypothesize** what might grow after incubation; **defer** detailed colony interpretation to Lab 17.

Reference: 14 species (TSA / TSB / slants)

These organisms are available in the teaching collection (growth on **tryptic soy** media). **Gram stain (typical)** is the reaction most teaching strains show; individual isolates or mislabeled stocks can vary—**always trust your stain or instructor's key** over this table if they disagree.

#	Species	Gram stain (typical)
1	*Serratia marcescens*	Negative
2	*Micrococcus luteus*	Positive
3	*Pseudomonas aeruginosa*	Negative
4	*Morganella morganii*	Negative
5	*Enterococcus faecalis*	Positive
6	*Citrobacter freundii*	Negative
7	*Staphylococcus aureus*	Positive
8	*Proteus vulgaris*	Negative
9	*Escherichia coli*	Negative
10	*Providencia alcalifaciens*	Negative
11	*Proteus mirabilis*	Negative
12	*Enterobacter aerogenes*	Negative
13	*Lactococcus lactis*	Positive
14	*Providencia stuartii*	Negative

Reference: Gram-positive vs. Gram-negative (structure and drug access)

Gram staining (crystal violet, iodine, alcohol decolorizer, safranin counterstain) reflects **peptidoglycan thickness** and **outer membrane** architecture—not virulence or "how dangerous" an organism is.

Feature	Gram-positive	Gram-negative
Peptidoglycan wall	Thick, exposed to the outside	Thin; sits between inner membrane and outer membrane
Outer membrane	Absent	Present (lipopolysaccharide, porins); acts as an extra permeability barrier
After staining	Purple/violet tends to remain	Alcohol extracts violet from thin wall; cells pick up **pink** counterstain

Why this matters for antibiotics: Drugs that must reach **cytoplasmic or periplasmic targets** (many **β-lactams** through porins, **aminoglycosides** that need active transport) are influenced by **outer-membrane permeability** and **efflux**. **Vancomycin** is a large **glycopeptide** that binds peptidoglycan **precursors**; it cannot cross the **intact outer membrane** of typical Gram-negative cells, so it is used clinically against **Gram-positive** pathogens. **Real cultures** also carry **resistance genes** (β-lactamases, altered targets, efflux), so a **missing zone** on a disc does not automatically mean "wrong Gram call"—interpret with your instructor.

Mechanism summaries in this lab align with standard reviews of antimicrobial targets (e.g., Kapoor et al., 2017, PMC5672523 — open access).

Reference: antibiotic discs available

Your group will **choose four** of the following for quadrants **A–D** on each strain plate. Use the **same four** across all three plates so you can compare results.

Gram-positive emphasis (G+):

- **Penicillin** — β-lactam; cell wall (PBPs). Many G– resist via β-lactamases / outer membrane.
- **Vancomycin** — glycopeptide; cell wall precursors. Too large to cross G– outer membrane.
- **Erythromycin** — macrolide; 50S ribosome. Porins/efflux limit most G– activity.

Gram-negative emphasis (G–):

- **Gentamicin** — aminoglycoside; 30S ribosome. G– aerobes often sensitive.
- **Amikacin** — aminoglycoside; 30S ribosome. Broader vs. some resistant G– strains.
- **Streptomycin** — aminoglycoside; 30S ribosome. Classic *M. tuberculosis* drug.
- **Nalidixic acid** — quinolone (1st gen.); DNA gyrase. G– emphasis.

Broad-spectrum (G+ and G–):

- **Ampicillin** — β-lactam; cell wall (PBPs). Broader than penicillin for some G– enterics; resistance common.
- **Ciprofloxacin** — fluoroquinolone; DNA gyrase. Good G– penetration; G+ activity varies.
- **Chloramphenicol** — amphenicol; 50S ribosome. Broad in principle; resistance common.
- **Tetracycline** — 30S ribosome. Broad bacteriostatic; efflux resistance common.
- **Sulfamethoxazole** — sulfonamide; folate synthesis. Broad in principle; resistance via bypass.

> ***Tip:*** *Pick antibiotics from* ***different categories*** *(e.g., one G+, one G–, one broad, one from a different target) for the most informative comparison.*

Safety and aseptic technique

- **BSL-1:** No eating or drinking; **gloves** when handling cultures and plates; wash hands after lab.
- **Bench: Clear and wash** your table; disinfect with **70% ethanol** (or as directed) before and after work.
- **Bunsen burner:** Tie back hair; know **emergency gas shutoff**; never leave flame unattended.
- **Hot glass / metal:** Flamed spreaders and tube rims stay **hot** briefly — do not set on plastic or burn yourself.
- **Incubation:** Store plates **inverted**; do not open plates after incubation except under instructor guidance (aerosols).

Materials (per station)

- Tryptic soy agar (TSA) plates
- **14** stock cultures in **TSB broth** and/or **TSA slants** (as posted)
- Sterile **droppers** (or Pasteur pipettes and bulbs)
- **Glass spreader** or bent **glass rod** (or disposable spreader, as provided)
- **70% ethanol** in beaker for dipping glass spreader between strains (if protocol uses shared spreader)

- **Bunsen burners**, striker
- **Antibiotic discs** (supply of agents listed above — each group chooses four)
- **Sharpie** for labeling **bottom** of plates (agar side)
- **Tape** for sealing take-home plates (Part B)

Part A — Plating and inoculating with antibiotic discs

A.1 Group assignments and chosen antibiotics

Group members: ____________________

Your group's four antibiotics:

- A = ____________________
- B = ____________________
- C = ____________________
- D = ____________________

Assigned strains and incubator log

#	Strain (exact label)	Gram status	Incubator temp / date in
1			
2			
3			
4			

A.2 Setup and prepare the Petri dishes

1. **Clear and wash** the bench; disinfect when dry.
2. **Put on gloves.**
3. Review **Bunsen burner** safety, then **light** your burner.

Your instructor will assign **three strains** to your group — ideally **at least one Gram-positive** and **at least one Gram-negative** species. You will prepare **one plate per strain** (three plates total). You will use the **same four antibiotics** listed in A.1 on every plate.

1. On the **bottom** of each TSA plate (the side with the agar), use a marker to draw **two perpendicular lines** through the center so the plate is divided into **four quadrants**.

2. Label the quadrants **A**, **B**, **C**, and **D** on the **bottom** (same side as the agar).
3. Write the **strain name**, **Gram status**, and your **group initials** on the **bottom** of each plate.

A.3 Hypothesize before plating

Before plating, **predict** whether each of your four antibiotics (**A–D**) will produce a zone of inhibition (effective) or no zone (ineffective) for each strain. Base your prediction on **Gram status** and **drug target**.

Strain 1: __________ (G+ / G–: __________)

- A: zone / no zone — because __________
- B: zone / no zone — because __________
- C: zone / no zone — because __________
- D: zone / no zone — because __________

Strain 2: __________ (G+ / G–: __________)

- A: zone / no zone — because __________
- B: zone / no zone — because __________
- C: zone / no zone — because __________
- D: zone / no zone — because __________

Strain 3: __________ (G+ / G–: __________)

- A: zone / no zone — because __________
- B: zone / no zone — because __________
- C: zone / no zone — because __________
- D: zone / no zone — because __________

A.4 Sterile transfer and lawn

1. **Alcohol and flame** the **glass spreader** (or rod) as demonstrated; allow to **cool** without touching non-sterile surfaces (e.g., rest on inside of sterile plate lid edge if that is your lab's method — **follow your instructor**).
2. **Gently mix** the assigned **broth tube** (light shake) to resuspend cells.
3. **Remove the cap carefully** without setting the inner cap on the bench.
4. Use a **sterile dropper** to withdraw broth; place **five drops** of the strain onto the **center** of the agar (or as directed). Replace the cap.
5. **Spread** the inoculum evenly over the **entire surface** of the plate to form a **lawn** (technique as demonstrated). If using one spreader for multiple steps, **re-sterilize** between strains.

A.5 Antibiotic discs (same four on every plate)

1. Using **sterile forceps** (dipped in alcohol and flamed, then cooled — or disposable forceps), place **one disc** into each quadrant (**A**, **B**, **C**, **D**) using the four antibiotics your group chose.
2. **Gently press** discs to contact agar if your instructor directs (do not smash).
3. **Invert** the plate, **tape** if required for transport, and place in the **designated incubator** (temperature as posted, often **37 °C**).
4. **Repeat** steps A.4–A.5 for each of your three assigned strains.

> ***After incubation:*** *Interpret zones, compare to* ***Gram category*** *and* ***drug class****, and complete air-plate colony work in* ***Lab 14 follow-up****.*

Part B — Air-exposure plates (take-home or classroom)

Your group has **four additional TSA plates** (one per person when possible).

1. **Label** the **bottom** of your plate with your **name**, **date**, and **location** where you will expose it (e.g., "kitchen counter," "car dashboard," or "CR classroom bench").
2. **Remove the lid** and expose the **open plate** to air in that space for **1 hour** (timer). **Do not** touch the agar with fingers or objects.
3. **Replace the lid**; **tape** the plate closed if you are transporting it.
4. If you **cannot** take a plate home, expose it in an **assigned area of the classroom** for 1 hour with instructor approval, then add it to the class incubator batch.

Part B — air exposure log

#	Student	Exposure location (be specific)	Start time	End time (1 hr)	Incubator date / temp
1					
2					
3					
4					
5					

Hypothesis: Before incubation, what **two** types of microbes (categories or examples) do you predict are most likely to appear on your air plate, and why?

After growth: *Describe colonies and compare locations in **Lab 14 follow-up**.*

Conclusion

Q1. What part of **aseptic technique** was easiest for you, and what will you **focus on improving** next time?

Q2. How does testing the **same four antibiotics** on each strain help you **compare susceptibility** across Gram-positive and Gram-negative bacteria?

Q3. Look back at your **hypothesis table** (section A.3). Which prediction do you feel **most confident** about, and why?

Q4. If a strain shows **no zone of inhibition** around a disc, does that automatically mean the organism is the wrong Gram type? What else could explain a missing zone?

Lab 14 — Microbiology. Tryptic soy media; 14 teaching strains; standardized 4-antibiotic disc susceptibility + environmental air sampling. Pair with Lab 14 follow-up after incubation.